

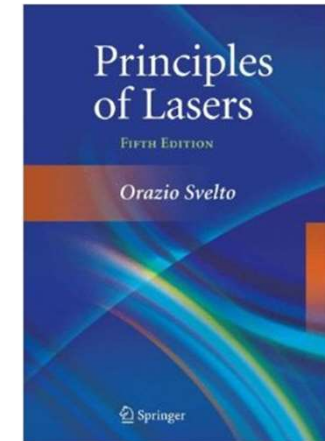


南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers

Ch2-3 Open Cavity & Gaussian Beam



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Chapter 2: Open Cavity & Gaussian Beam

❖ 2.3 Mode for Open Cavity & Diffraction Theory

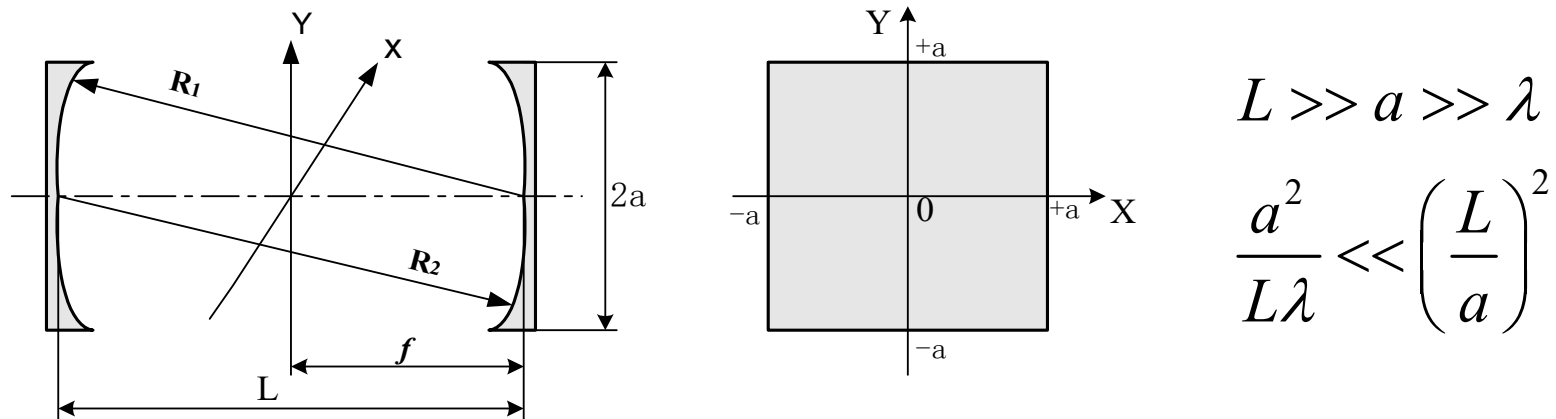
- Introduction
- The self-reconstruction of mode
- Fresnel Kirchhoff's law
- The physical meaning of the integral equation

❖ 2.5 The self-reconstructed mode of square confocal cavity

- Variables separation
- Solutions: analytical solution and approximation solution

Type of Symmetrical Confocal Cavity	Analytical Solution	Approximation
Square	Prolate-ellipsoid function	Hermite-Gaussian function
Circular	Hyper-ellipsoid function	Laguerre-Gaussian function

2.5 Self-reconstructed mode of square confocal cavity



$$V(x, y) = \gamma \frac{i}{\lambda L} \iint_s V(x', y') e^{-ik\rho(x, y, x', y')} ds'$$



$$\sigma_m F_m(X) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_m(X') e^{iXX'} dX'$$

$$\sigma_n F_n(Y) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_n(Y') e^{iYY'} dY'$$

$$\sigma_m F_m(X) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_m(X') e^{iXX'} dX'$$

$$\sigma_n F_n(Y) = \sqrt{\frac{ie^{-ikL}}{2\pi}} \int_{-\sqrt{c}}^{+\sqrt{c}} F_n(Y') e^{iYY'} dY'$$

Analytical solution: Prolate-ellipsoid function (长椭球函数系)

$$2i^m \underbrace{R_{0m}^{(1)}(c,1)}_{\text{Radial prolate-ellipsoid function}} \underbrace{S_{0m}^{(1)}(c,t)}_{\text{Angular prolate-ellipsoid function}} = \int_{-1}^{+1} e^{ictt'} S_{0m}(c,t') dt' \quad (m=0, 1, 2, \dots)$$

Radial prolate-ellipsoid function Angular prolate-ellipsoid function
 径向长椭球函数 角向长椭球函数

• **Eigen function** — Angular Amplitude & Phase distribution

$$V_{mn}(x, y) = F_m(X) G_n(Y) = S_{0m}\left(c, \frac{X}{\sqrt{c}}\right) S_{0n}\left(c, \frac{Y}{\sqrt{c}}\right)$$

$$F_m(X) = S_{0m}\left(c, \frac{X}{\sqrt{c}}\right) = S_{0m}\left(c, \frac{x}{a}\right) \quad G_n(Y) = S_{0n}\left(c, \frac{Y}{\sqrt{c}}\right) = S_{0n}\left(c, \frac{y}{a}\right)$$

• Eigen value—Radial

Phase shift & loss of the mode

$$\begin{aligned}
 \sigma_m \sigma_n &= \chi_m \chi_n i e^{-ikL} \\
 \chi_m &= \sqrt{\frac{2c}{\pi}} i^m R_{0m}^{(1)}(c, 1) \\
 \chi_n &= \sqrt{\frac{2c}{\pi}} i^n R_{0n}^{(1)}(c, 1)
 \end{aligned}
 \quad \rightarrow \quad
 \begin{aligned}
 \sigma_m &= 2i^m \sqrt{c} R_{0m}^{(1)}(c, 1) \left(\frac{i e^{-ikL}}{2\pi} \right)^{1/2} \\
 \sigma_n &= 2i^n \sqrt{c} R_{0n}^{(1)}(c, 1) \left(\frac{i e^{-ikL}}{2\pi} \right)^{1/2}
 \end{aligned}$$

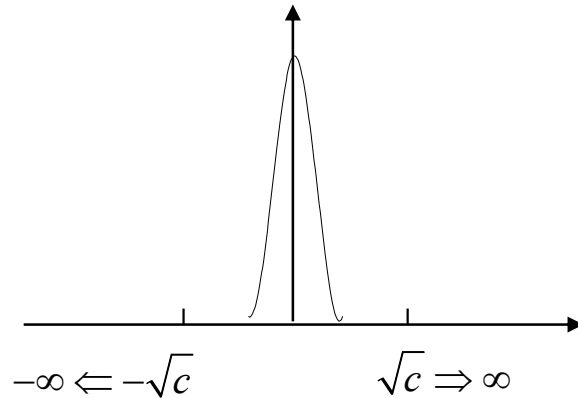
$$\gamma_{mn} = \frac{1}{\sigma_m \sigma_n} \quad \sigma_m \sigma_n = 4N e^{-i \left[kL - (m+n+1) \frac{\pi}{2} \right]} R_{0m}^{(1)}(c, 1) R_{0n}^{(1)}(c, 1)$$

$$R_{0m}^{(1)}(c, 1), R_{0n}^{(1)}(c, 1) \quad \text{Real function}$$

For specific c ($c=2\pi N=2\pi a^2/L\lambda$), when m and n is integer, we can obtain a series of eigen functions, which describe the amplitude and phase distribution on the confocal cavity mirror; Meanwhile, we can obtain a series of eigen values, which describe the phase shift and loss of the mode. **TEM_{mn}**

Hermite-Gaussian function: approximation

Approximation for prolate-ellipsoid function (长椭圆函数)



Approximation requirement:

Mode distribute near the center of the mirror

$x, y \ll a$ 菲涅尔数较大 $c = 2\pi N \gg 1$

$$\sigma_m F_m(X) = \left(\frac{ie^{-ikL}}{2\pi} \right)^{1/2} \int_{-\sqrt{c}}^{+\sqrt{c}} F_m(X') e^{iXX'} dX' \quad \int_{-\infty}^{+\infty} \dots$$

$$\frac{d^2 F_m(X)}{dX^2} + [\lambda - X^2] F_m(X) = 0 \quad \text{Schrödinger equation of linear harmonic oscillator}$$

for $X \rightarrow \infty$, $F_m(X) \rightarrow 0$, the solution is Hermite-Gaussian function

Angular
prolate-ellipsoid function → Hermit polynomials Gaussian function

$$F_m(X) = S_{0m} \left(c, \frac{X}{\sqrt{c}} \right) = C_m \underbrace{H_m(X)}_{\text{Hermit polynomials}} e^{-\frac{X^2}{2}} \quad \leftarrow \text{Gaussian function}$$

$$G_n(Y) = S_{0n} \left(c, \frac{Y}{\sqrt{c}} \right) = C_n H_n(Y) e^{-\frac{Y^2}{2}}$$

Eigen Function ↓ $X, Y \rightarrow x, y$ (镜面上直角坐标)

$$V_{mn}(x, y) = C_{mn} H_m \left(\frac{\sqrt{c}}{a} x \right) H_n \left(\frac{\sqrt{c}}{a} y \right) e^{-\frac{c}{2a^2}(x^2 + y^2)}$$

$$= C_{mn} H_m \left(\sqrt{\frac{2\pi}{L\lambda}} x \right) H_n \left(\sqrt{\frac{2\pi}{L\lambda}} y \right) e^{-\frac{x^2 + y^2}{(L\lambda/\pi)}}$$

C_{mn} — constant. We can use this approximation to discuss the field distribution on the confocal cavity's mirror

Hermite Polynomials & Hermite-Gaussian Function

1) m -order Hermite polynomial

$$H_m(X) = (-1)^m e^{X^2} \frac{d^m}{dX^m} (e^{-X^2}) = m \sum_{k=0}^{\left[\frac{m}{2}\right]} \frac{(-1)^k m!}{k! (m-2k)!} (2X)^{m-2k} \quad \text{where } (m = 0, 1, 2, \dots)$$

Take integer part 取整数部分

2) The first several order of Hermite polynomial

$$H_0(X) = 1$$

$$H_1(X) = 2X$$

$$H_2(X) = 4X^2 - 2$$

$$H_3(X) = 8X^3 - 12X$$

$$H_4(X) = 16X^4 - 48X^2 + 12$$

3) The zero position of the m -order Hermite polynomial: $H_m(X)=0$ has m solutions.

Therefore, the m -order of hermite polynomial has m zero positions (along the x axis, the position of the light that equal to zero has m positions).

4) The m -order Hermite-Gaussian function

$$H_m(X) e^{-X^2/2} \quad H_n(Y) e^{-Y^2/2} \Rightarrow V_{mn}(x, y) = C_{mn} H_m\left(\frac{\sqrt{c}}{a} x\right) H_n\left(\frac{\sqrt{c}}{a} y\right) e^{-\frac{c}{2a^2}(x^2+y^2)}$$

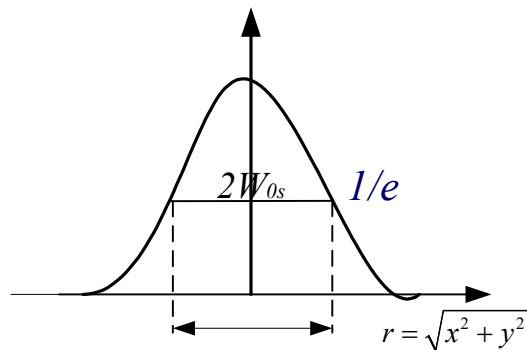
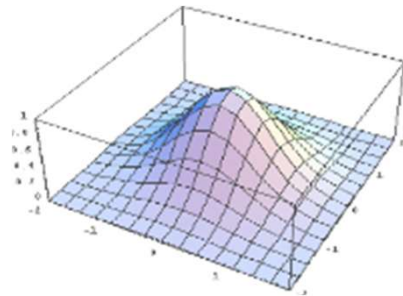
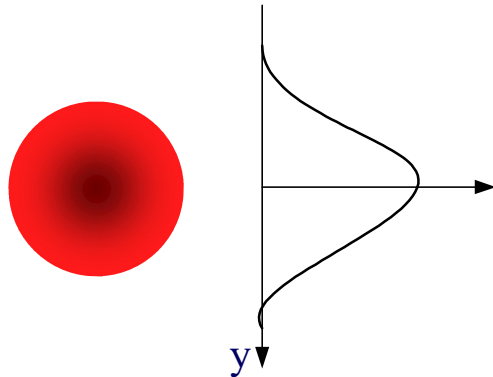
The field distribution on the mirror: determine by m and n , called transverse mode order

Transverse mode: Field distribution perpendicular to the direction of propagation

determine by m and n

Amplitude distribution of the field on the mirror

Low order fundamental transverse mode TEM₀₀ (m=n=0)



$$V_{mn} = C_{mn} H_m \left(\frac{\sqrt{c}}{a} x \right) H_n \left(\frac{\sqrt{c}}{a} y \right) e^{-\frac{c}{2a^2}(x^2+y^2)}$$

$$H_0(x) = H_0(y) = 1$$

TEM₀₀ is Gaussian shape
基模在镜面上分布为高斯型

$$V_{00} = C_{00} e^{-\frac{x^2+y^2}{L\lambda/\pi}} = C_{00} e^{-r^2/w_{0s}^2}$$

The radius of the light spot w_{0s} : amplitude decrease to $1/e$ of the maximum.

$$V_{00} = C_{00} e^{-\frac{x^2+y^2}{L\lambda/\pi}} = C_{00} \frac{1}{e} \quad x^2 + y^2 = L\lambda/\pi$$

$$w_{0s} = \sqrt{x^2 + y^2} = \sqrt{L\lambda/\pi} \quad \text{Only related to } L, \text{ not related to the size of the mirror}$$

For CO₂ gas laser, $L=1$ m, $\lambda=10.6$ μm , $w_{0s}=1.84$ mm

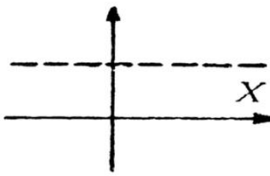
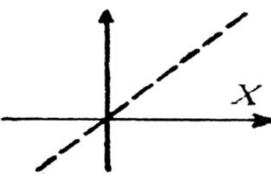
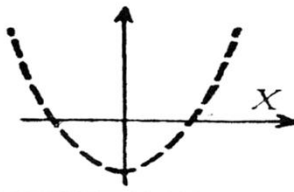
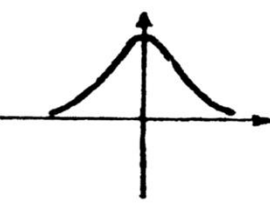
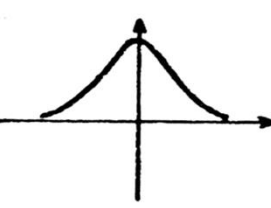
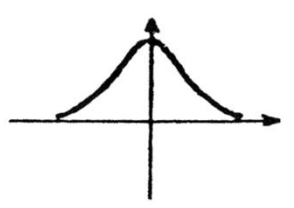
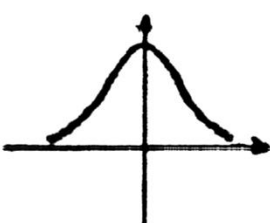
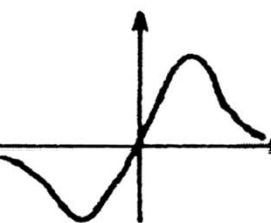
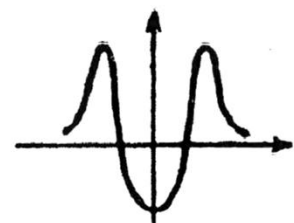
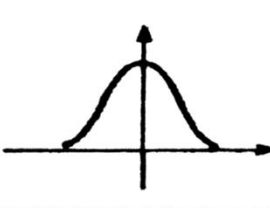
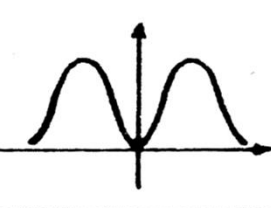
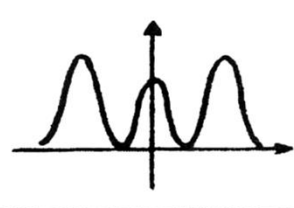
HeNe gas laser, $L=30$ cm, $\lambda=0.6328$ μm , $w_{0s}=0.25$ mm

Amplitude distribution of the field on the mirror

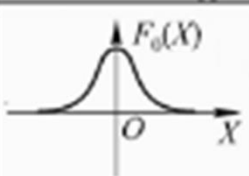
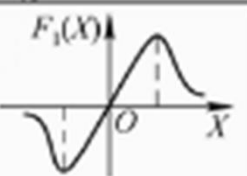
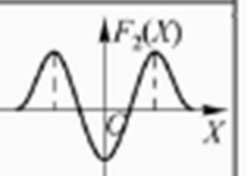
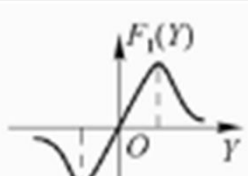
High order transverse mode TEM_{mn} (m and $n \neq 0$ simultaneously)

$$\begin{aligned} V_{mn}(x, y) &= C_{mn} H_m \left(\frac{\sqrt{c}}{a} x \right) H_n \left(\frac{\sqrt{c}}{a} y \right) e^{-\frac{c}{2a^2}(x^2+y^2)} \\ &= C_{mn} H_m \left(\sqrt{\frac{2\pi}{L\lambda}} x \right) H_n \left(\sqrt{\frac{2\pi}{L\lambda}} y \right) e^{-\frac{x^2+y^2}{(L\lambda/\pi)}} \end{aligned}$$

$$\begin{aligned} H_0(X) &= 1, & H_1(X) &= 2X, & H_2(X) &= 4X^2 - 2... \\ V_{10}(x, y) &= C'_{10} x e^{-\frac{x^2+y^2}{w_{0s}^2}}, & V_{01}(x, y) &= C'_{01} y e^{-\frac{x^2+y^2}{w_{0s}^2}} \\ V_{20}(x, y) &= C'_{20} (4x^2 - w_{0s}^2) e^{-\frac{x^2+y^2}{w_{0s}^2}}, & V_{11}(x, y) &= C'_{11} xy e^{-\frac{x^2+y^2}{w_{0s}^2}} \\ &... & & \end{aligned}$$

	$m=0$	$m=1$	$m=2$
$H_m(X)$			
$F = e^{-\frac{x^2+y^2}{w^2}}$			
$FH_m(X)$			
$I_\infty F^2 H_m^2(X)$			
TEM_{m0}			

The zero point of Hermite polynomials determines the pitch line of the field
厄米多项式的零点决定场的节线

TEM_{nm} 光斑图像 $F_n(Y)$ \ $F_m(X)$		$m=0$	$m=1$	$m=2$
				
$n=0$		TEM_{00} 	TEM_{10} 	TEM_{20} 
		TEM_{01} 	TEM_{11} 	TEM_{21} 



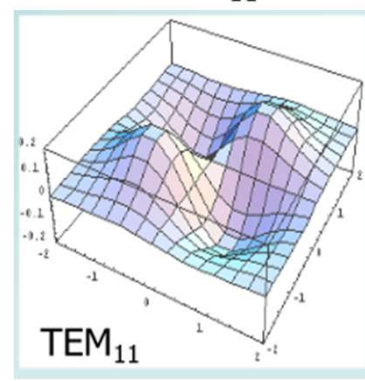
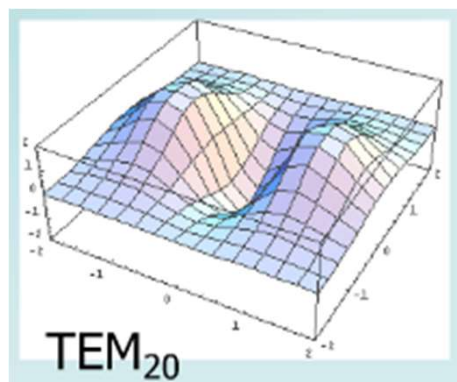
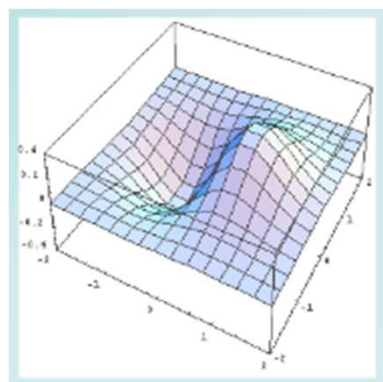
TEM_{10}



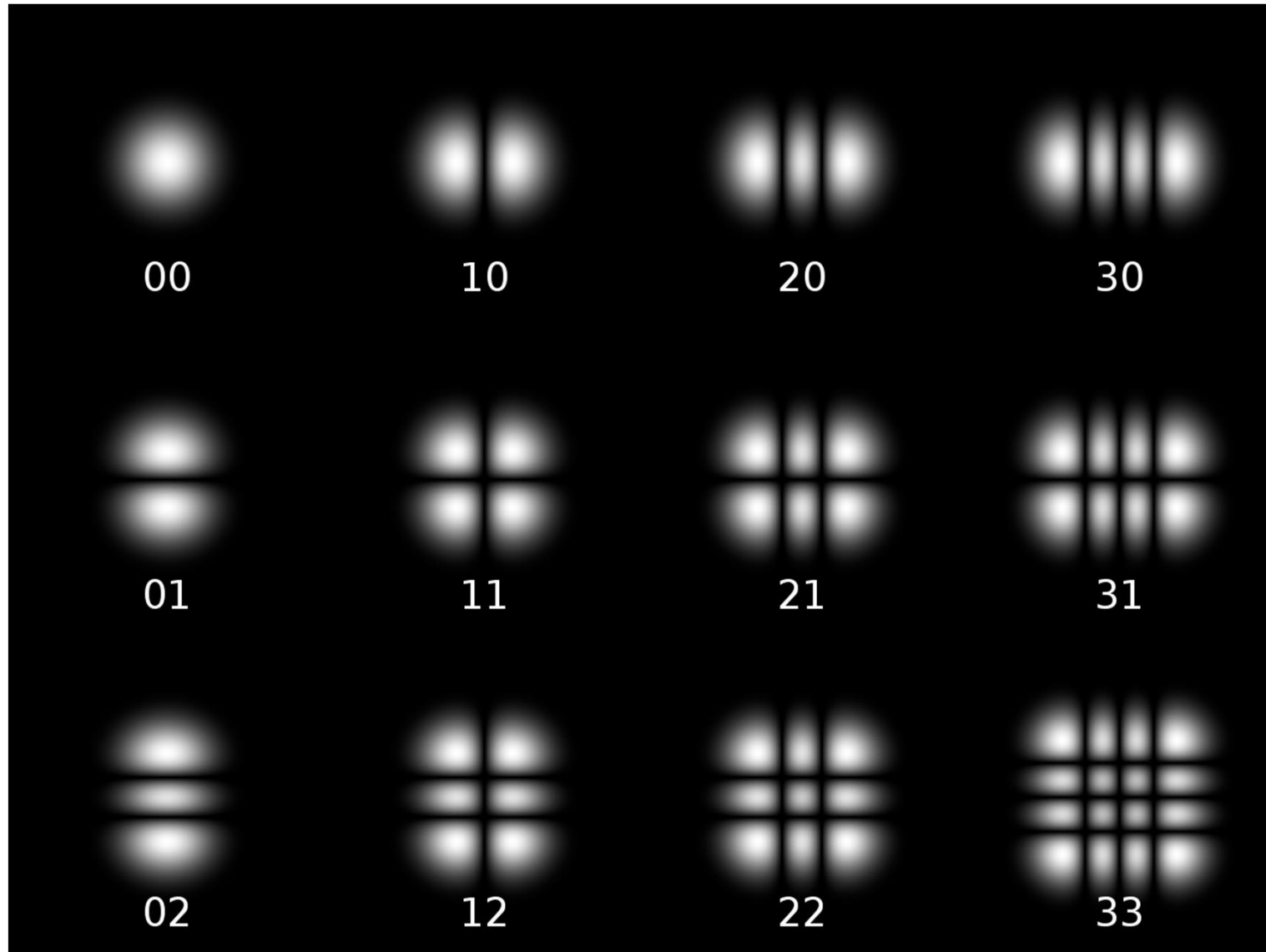
TEM_{20}



TEM_{11}



Intensity distribution



- The high order transverse mode is more complex than the fundamental base mode. They have pitch line and with bigger spot
- The spot size of the high order mode is bigger

According to ISO, the square of spot radius is defined as four times of the mean square difference of light field distribution

根据国际标准化组织（ISO）光斑半径的平方定义为光场分布均方差值的四倍

$$w_{ms}^2 = \frac{4 \int_{-\infty}^{+\infty} F_m(X) (x - \bar{x})^2 F_m(X) dX}{\int_{-\infty}^{+\infty} [F_m(X)]^2 dX}$$

$$w_{ns}^2 = \frac{4 \int_{-\infty}^{+\infty} G_n(Y) (y - \bar{y})^2 G_n(Y) dY}{\int_{-\infty}^{+\infty} [G_n(Y)]^2 dY}$$

$w_{ms}^2 = (2m + 1) w_{0s}^2$ $w_{ns}^2 = (2n + 1) w_{0s}^2$		$\frac{w_{ms}}{w_{0s}} = \sqrt{2m + 1}, \quad \frac{w_{ns}}{w_{0s}} = \sqrt{2n + 1}$
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Diffraction Loss per reflection --- Determined by N

$$\delta_{mn} = 1 - \left| \frac{1}{\gamma_{mn}} \right|^2 = 1 - |\sigma_m \sigma_n|^2 \stackrel{\sigma_m \sigma_n = \chi_m \chi_n e^{-ikL}}{=} 1 - |\chi_m \chi_n|^2$$

Radial prolate-ellipsoid function 径向长椭球函数

$$\chi_m = \sqrt{2c/\pi} i^m R_{0m}^{(1)}(c, 1)$$

$$\chi_n = \sqrt{2c/\pi} i^n R_{0n}^{(1)}(c, 1)$$

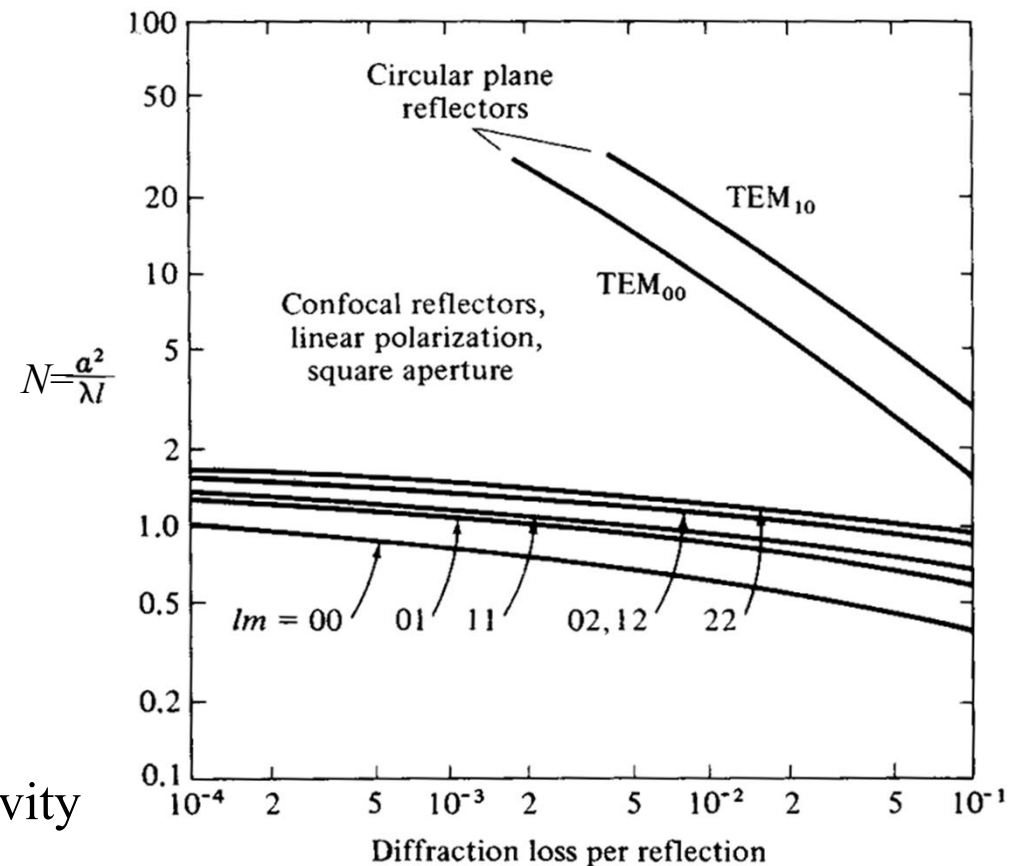
δ_{mn} is closely-related to N

Inside the cavity

$$N \uparrow \rightarrow \delta_D \downarrow, \quad m, n \uparrow \rightarrow \delta_D \uparrow \quad \delta_{01} > \delta_{00}$$

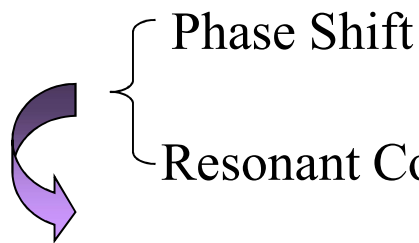
Different kind of cavities

Confocal cavity < parallel planar cavity



Phase Shift & Resonant Frequency

$$\sigma_m \sigma_n = 4Ne^{-i\left[\frac{kL-(m+n+1)\frac{\pi}{2}}{2}\right]} R_{0m}^{(1)}(c,1) R_{0n}^{(1)}(c,1)$$



Phase Shift

$$\delta\varphi_{mn} = -kL + (m+n+1)\frac{\pi}{2} = -kL + \Delta\varphi_{mn}$$

Resonant Condition

$$2\delta\varphi_{mn} = -q2\pi$$

Resonant frequency

$$\nu_{mnq} = \frac{c}{2L'} \left[q + \frac{1}{2}(m+n+1) \right] \Rightarrow \nu_{00q} \approx \frac{c}{2L'} q$$

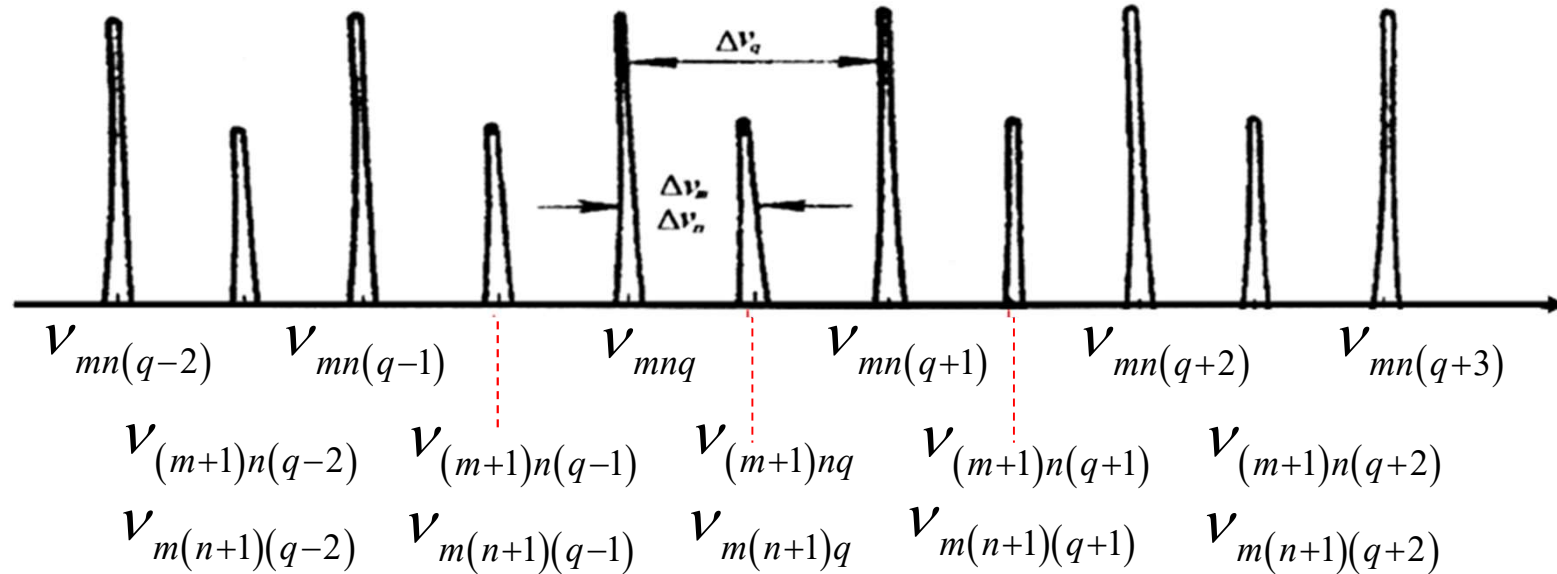
Mod spacing between adjacent longitudinal mode

$$\Delta\nu_q = \nu_{mn(q+1)} - \nu_{mnq} = \frac{c}{2L'} = \frac{c}{2\eta L}$$

For the same longitudinal mode, the mode spacing between transverse mode

$$\Delta\nu_n = \nu_{m(n+1)q} - \nu_{mnq} = \frac{c}{4\eta L} = \frac{1}{2} \Delta\nu_q = \Delta\nu_m$$

The change of m , n , q lead to the change of resonant frequency, with the same order of magnitude



❖ Multiple modes exist at one frequency

$$V_{mnq} = V_{(m-1)(n+1)q} = V_{m(n-2)(q+1)} = V_{(m-2)n(q+1)} = V_{(m+1)(n-3)(q+1)} = \dots$$

$$V_{(m+1)nq} = V_{m(n+1)q} \quad V_{(m+1)n(q+1)} = V_{m(n+1)(q+1)} \quad \dots$$

- Mode degeneracy: modes number with the same frequency $\Delta\nu_n = \frac{c}{4\eta L} = \frac{1}{2} \Delta\nu_q$
- Mode degeneracy is a characteristic of confocal cavity
- The single reflection loss of modes with the same frequency $\Delta\nu_m = \frac{1}{2} \Delta\nu_q$ is not the same